How the 2015–16 El Niño developed

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Abstract

The 2015–16 El Niño peaked at +2.25 on NOAA's Relative Oceanic Niño Index (RONI) in NDJ 2015–16 and at +2.59 on the traditional ONI: on the relative index it is the weakest of the three super events in this series, on the traditional index the strongest, and within observational uncertainty the three cannot be separated [3] [14]. Its development was a two-year story. A recharged Pacific and strong westerly bursts in early 2014 set up a widely forecast major event that stalled: westerly bursts stopped from April to June, a basin-wide easterly burst in June knocked the Niño indices back to normal, and a cold south-eastern Pacific diverted the winds [16] [17] [20]. The same easterly burst stopped the warm water from draining away, leaving the basin charged for 2015 [18]. From March 2015 a strong burst, then an unusually long run of bursts through the year — the most in any year of the 1979–2016 reanalysis — drove four downwelling Kelvin waves and coupled growth to a peak in November–January [13] [6]. The event was centred further west than 1982–83 and 1997–98, with record Niño 4 and weaker warming at the South American coast, consistent with an added role for central-Pacific, subtropically forced dynamics; a North Pacific coupled mode linked to the 2013–15 'Blob' is a proposed Northern Hemisphere trigger [12]

[15] [27]. TAO mooring reporting declined sharply in 2012–14, so the failed 2014 onset was observed with far fewer moorings than 1997 [35] [5].

Main findings

1. **Precondition (grade A−; high confidence).** Warm water volume stayed positive from mid-2013; the June 2014 easterly burst prevented and then reversed the discharge that a 2014 El Niño would have caused, giving 2015 a head start [3] [18].
2. **A failed trigger in 2014 (grade A; high confidence).** Two strong bursts in January–March 2014, then none from April to June; a wind-forced ocean model reproduces both the 2014 stall and the 2015 growth from bursts and easterly surges alone [16] [19] [6].
3. **Suppressors in 2014 (grade A for the easterly burst, B/C for the meridional gradient).** A basin-wide easterly burst in June and a south-eastern Pacific cold anomaly that drove meridional rather than zonal winds [17] [20].
4. **Trigger and growth in 2015 (grade A−; high confidence).** A strong burst in March 2015 and a record number of bursts through the year drove four downwelling Kelvin waves from April to November [6] [13] [5].
5. **Pattern (grade B; medium–high confidence).** Westward-centred warming, record Niño 4 and a weak coastal thermocline anomaly: eastern-Pacific dynamics plus stronger central-Pacific dynamics [13] [15] [12].
6. **North Pacific link (grade B/C; medium–low confidence).** A Victoria-mode/meridional-mode pathway from the North Pacific is proposed for this event; it rests on one statistical study and general hindcasts [27] [23].
7. **Background warming (grade C).** The index adjustment lowers the peak by 0.34 °C, the most of the three events; one review attributes the extremity partly to background warming, but no event attribution is made here [2] [11].

1. Introduction

Early in 2014 the tropical Pacific looked much as it had in early 1997: a recharged equatorial heat content, an eastward-displaced warm pool and strong westerly bursts. Many models forecast a strong El Niño, some an extreme one [16] [17]. It did not happen in 2014; it happened, at extreme strength, in 2015. That makes 2015–16 the only one of the three historical super events with a documented false start, and a natural test of what decides whether a charged Pacific turns into a major El Niño. The forecasts for 2015 itself were good, in contrast to the 'overexuberant' predictions of the year before [12].

This paper reconstructs the sequence from mid-2013 to the La Niña of late 2016 with the same framework, evidence grades and primary index (RONI) as Papers 1 and 2, and with

identical reanalysis diagnostics, so that the three events can be compared.

2. Definitions, data and methods

Index. RONI is the primary index [1] [2]; ONI is shown alongside. The choice matters more for this event than for the others: removing the tropical-mean warming lowers the peak from +2.59 to +2.25.

Observations. Argo floats and satellite altimetry, winds and salinity were all available. The TAO/TRITON array was not: its reporting declined dramatically from mid-2012 to early 2014 [35]. In the moorings used here, daily equatorial wind records exist for 33–41% of site-days from July 2013 to March 2014 and 36% in April–June 2014, against 83–97% from October 2014 to September 2015 [5].

Reanalysis diagnostics. The burst and MJO diagnostics of Papers 1 and 2 were re-run for July 2013 – December 2016 with the same definitions: 1000 hPa anomaly of at least 5 m s^{-1} for at least 3 days over at least 10° of longitude, 5°S–5°N, 120°E–150°W, against a 1979–2000 climatology, with the MJO index patterns fitted to 1979–2001 [6]. Because easterly bursts are broad rather than intense, they are described with the 150°E–100°W mean anomaly instead of the burst rule. Kelvin waves are traced in TAO 20 °C isotherm depth where the moorings reported.

Evidence grades are those of Papers 1 and 2 (A: event observation plus a model or budget study isolating the effect; B: event observation with a plausible mechanism; C: correlation, composite or theory; D: speculative or untested). Several key studies were read only as abstracts; the reference list records what was read.

3. Pre-event conditions and the false start of 2014

A charged Pacific. After weak cool conditions in 2013 (RONI −0.48 in JJA 2013), warm water volume was positive throughout July 2013 – February 2014 (+0.78 to +1.25 × 10^{14} m^3) [3] [7]. In winter 2013–14 an exceptional warm anomaly — the 'Blob' — formed in the north-eastern Pacific, the largest there since at least the 1980s, through reduced ocean heat loss and weak cold advection under persistently high pressure [25].

Early 2014: the onset that stalled. The reanalysis shows two strong westerly bursts, on 24 Jan 2014 – 29 Jan 2014 (+9.5 m s^{-1} at 145°E) and 19 Feb 2014 – 5 Mar 2014 (+9.2 m s^{-1} at 155°E), and then none until a weak one in August; 2014 has 3 bursts in all [6]. Heat content east of the Date Line rose from −0.33 °C in January to +1.60 °C in March and warm water volume peaked at +2.05 × 10^{14} m^3 [3]. Until April the warm pool was displaced east as at the onset of strong events, but by July it had retreated: in ocean simulations the difference from 1997 is the absence of westerly events from April to June [16]. A wind-forced model

built from the available buoy winds explains the early development and mid-year reversal by a springtime surplus of westerly events followed by a summer balance between westerly events and easterly surges [19].

Suppressors. In June an unusually strong, basin-wide easterly burst preceded a rapid fall of all major Niño indices to near normal; afterwards two separate warm centres persisted in the central and eastern Pacific and the Bjerknes feedback did not engage [17]. In the reanalysis this easterly burst is broad rather than intense: the 150°E–100°W mean 1000 hPa anomaly reaches -1.94 m s^{-1} on 11 Jun 2014, the 5th-lowest of the 1280 days analysed, yet no 5 m s^{-1} easterly burst meets the burst rule [6]. A south-eastern Pacific cold anomaly with warmth to the north drove strong meridional winds near the equator rather than the zonal winds that growth needs; in forecast experiments it accounts for about 40% of the 2014 amplitude error [20].

The head start. The same easterly burst prevented, and then reversed, the discharge of equatorial heat content that a 2014 El Niño would have caused; warm water volume rebuilt to +1.49 and +1.57 × 10^{14} m^3 in August–September 2014 [18] [3]. Weak warm conditions persisted (RONI +0.62 in NDJ 2014–15; ONI ≥ +0.5 from SON 2014), but by conventional definitions the 2014 event failed to qualify as an El Niño [18].

4. Development, March – December 2015

Trigger. The decisive burst came on 4 Mar 2015 – 16 Mar 2015: +12.4 m s^{-1} at 167.5°E, spanning 140°–175°E, during an active MJO (OMI 2.0) [6]. The restored TAO array traced its downwelling Kelvin wave from 165°E on 11 Mar 2015 to 110°W on 12 Apr 2015, 32 days at about 3.4 m s^{-1}; later waves left 165°E on 11 May 2015 and 4 Jul 2015 [5]. From April to November four downwelling Kelvin waves initiated and enhanced the eastern-central warming, and eastward current anomalies advected the warm pool east in spring and summer [13]. A freshwater anomaly in the equatorial Pacific larger than anything in the 2004–2014 Argo record raised steric height in the west and increased eastward surface acceleration at the salinity front [32].

Coupling. RONI rose from +0.89 (AMJ) to +1.27 (JJA), with SOI −0.6 to −1.1 and Niño 1+2 reaching +2.48 °C in July; public El Niño advisories were issued in the first half of 2015 [3] [12]. Unlike 1997, when most bursts preceded coupling, bursts kept coming as the event grew: 12 from late June to October, several centred east of the Date Line, and 17 in calendar 2015 — the most of any year in 1979–2016 (1997 had 16, 1982 had 13; the 1979–2001 mean is 4.2) [6]. A wind-forced ocean model attributes the rapid 2015 warming to this abundance of westerly events [19].

Peak. RONI reached +2.03 (SON), +2.18 (OND) and +2.25 (NDJ); central-Pacific OLR fell to −49.9 W m^{-2} in December [3]. Niño 4 set a record and Niño 3 matched the earlier super

events, but Niño 1+2 was much weaker; enhanced convection lay about 20° further west, and the maximum thermocline anomaly near South America was only a third to a half of that in 1982 and 1997 [13]. Unlike previous major El Niños, SST anomalies were remarkably high in the western and central equatorial Pacific and milder near the coast of South America [12]. The 1997–98 event followed eastern-Pacific dynamics based on basin-wide thermocline changes; 2015–16 added central-Pacific dynamics dependent on subtropical forcing [15] [34].

5. Termination and transition, November 2015 – 2016

An upwelling Kelvin wave emerged in early November 2015 and led to a rapid decline [13]; warm water volume turned negative in December (-0.39×10^{14} m³) and fell to -3.36×10^{14} m³ by April 2016 [3]. RONI fell from +1.81 (JFM) to +0.08 (AMJ) and the event ended in May 2016 [3] [12]. During decay the SST anomalies lingered around the Date Line instead of retreating to the South American coast as in 1997–98 [15]. The southward shift of westerly anomalies in boreal winter, tied to the seasonal strengthening of the South Pacific Convergence Zone, is a key element in the termination of strong El Niño events in general [31]; its role in 2016 was not isolated in the studies read here. A weak La Niña followed (RONI −0.96 in SON 2016), promoted by a strong negative Indian Ocean Dipole that developed mainly from Indian Ocean conditions [28].

6. Interactions with the broader climate system

Intraseasonal variability. The MJO was active (OLR MJO index ≥ 1) on 3 Mar 2015 – 2 Apr 2015 and 29 May 2015 – 22 Jul 2015 (maximum 2.89), framing the March and late-June bursts [6]. Intraseasonal OLR variance at 160°E–150°W in October 2015 – March 2016 was 303 $W^2\ m^{-4}$, nearly twice that of October 1996 – March 1997 (166), as convection shifted east with the warm water [6]. In a model, El Niño-like SST shifts the MJO onto the equator and strengthens it in the central Pacific, favouring bursts [22]; a comparison of the three super events finds 2015–16 alone kept a prominent eastward-propagating MJO in its mature stage [10]. MJO events warm the waveguide only when they contain westerly bursts [8]. Every El Niño of 1971–2010 was accompanied by clustered westerly bursts, and whether they produce a central- or eastern-Pacific event depends on their timing within the recharge–discharge cycle [21]; in 2015 they fell on a basin already recharged by the aborted 2014 event.

North Pacific. The Blob persisted for several years and the PDO was positive from 2014 (up to +1.86 in December 2014) [26] [3]. One analysis identifies the Blob as part of an enhanced Victoria-mode coupled pattern forced by a North Pacific Oscillation-like circulation from 2013, a precursor of ENSO that bridged to the 2015–16 El Niño, which it

describes as triggered from the Northern Hemisphere, unlike other large events [27]. For the earlier super events a Southern Hemisphere circulation 'booster' had been identified [24]. Hindcast experiments find that the North Pacific Oscillation-induced meridional mode influences ENSO evolution more lastingly than a March MJO [23]. The evidence for 2015 itself remains statistical (grade B/C).

South Pacific. The 2014 south-eastern Pacific cold anomaly acted as a suppressor (above) [20].

Indian Ocean. The dipole index was weakly positive in August–October 2015 (CPC DMI +0.42 to +0.47 °C) [3]; IOD phase is largely set by ENSO through combination-mode wind and heat-flux forcing [29]. The strong negative IOD of 2016 mattered for the transition: without it the 2016 La Niña may have been substantially weaker [28].

Volcanic forcing. The tropical eruptions of VEI 4 in the period were Sinabung (September 2013, 3°N), Kelud (February 2014, 8°S) and Wolf (May 2015, on the equator); none was larger [4]. No study read here links them to the 2014–16 evolution (not a supported candidate).

Background warming. The tropical-mean adjustment lowers the peak by 0.34 °C (ONI +2.59, RONI +2.25), the largest difference of the three events [3] [2]. A review attributes the extremity of the event in part to the unusually warm 2014 conditions and to long-term background warming [11]; the rapid global temperature rise of 2014–16 has been explained by heat release from these El Niño conditions on top of the warming trend [33]. With SST uncertainty the three strongest events are not separable [14]. No event-specific attribution of amplitude is made here.

7. Causal synthesis and competing explanations

The most plausible sequence is: (1) the equatorial Pacific was recharged from 2013; (2) early-2014 bursts started an onset that stalled without spring bursts, under a June easterly burst and a cold south-eastern Pacific; (3) the easterly burst kept the heat in the basin; (4) a strong burst in March 2015 and a record run of bursts through 2015 drove repeated Kelvin waves and coupled growth; (5) central-Pacific dynamics and a North Pacific precursor gave the event its westward-shifted pattern; (6) an upwelling Kelvin wave from November 2015 and discharge ended it by May 2016.

Competing views concern weights rather than ingredients: whether the 2014 stall was mainly the missing spring bursts [16], the easterly burst [17], a balance of both [19] or the meridional SST gradient [20]; and whether the North Pacific precursor was a cause or a parallel response to the same atmospheric forcing [27]. Many 2015 bursts came after coupling had begun and are probably partly a response to it [9] [22].

8. What was forecast at the time

The predictions made in 2014 were 'overexuberant': models anticipated a strong event that did not develop [12]. A climate model with superimposed wind bursts suggests inherent limits to predictability once an easterly burst of this kind occurs [17]. In 2015 the life cycle and magnitude were well predicted by most models used by national meteorological services [12].

9. Uncertainties

- TAO data return in 2014 was poor, so wind-forced reconstructions of the 2014 stall depend on gap-filled winds [19] [5].
- The burst count uses a 1979–2000 climatology; any trend in the background winds since then affects counts in 2014–16.
- Several key studies (Xue and Kumar; Tseng et al.; Gasparin and Roemmich; Lim and Hendon) were read only as abstracts.
- The North Pacific link rests mainly on one statistical analysis.

10. Conclusion

The 2015–16 El Niño was built over two years. The ocean was ready in 2014, but the winds were not: without spring bursts, and with a strong June easterly burst and a cold south-eastern Pacific, the 2014 onset stalled. That failure kept the heat in place, and in 2015 a sustained, record run of westerly bursts turned it into an extreme, westward-shifted El Niño with a weaker coastal signature than 1982–83 or 1997–98. Compared with those events the trigger is again the westerly bursts, the precondition again the recharge — and what differs is the path from one to the other.

Evidence base: scope, index and data

Primary index: NOAA's Relative Oceanic Niño Index (RONI) [1] [2]. On RONI the 2015–16 peak is +2.25 (NDJ 2015), the weakest of the three historical events in this series (1982–83 +2.40, 1997–98 +2.28); on the traditional ONI it is +2.59, the strongest. The difference (0.34 °C) is the tropical-mean warming that RONI removes. Within SST uncertainty the three events cannot be ranked [14].

Observing system. Argo floats (from the 2000s) and satellite altimetry, salinity and winds were available, but TAO mooring data return had declined sharply from mid-2012 to early 2014, so the 2014 onset attempt is less well observed at the equator than 1997 [35] [19] [36].

Monthly indices, Jul 2013 – Dec 2016

Values from NOAA CPC, PMEL and NCEI files as held in the site's data archive; read from the source files, not retyped.

Month	Season	RONI	ONI	rel. Niño 3.4 (OISST)	Niño 3.4 (OISST)	Niño 1+2 (OISST)	Niño 4 (OISST)	SOI	W Pac trades	C Pac trades
2013-07	JJA 2013	-0.48	-0.45	-0.21	-0.26	-1.81	-0.17	0.8	0.6	-0.0
2013-08	JAS 2013	-0.38	-0.34	-0.49	-0.47	-1.03	-0.19	0.2	0.5	-0.4
2013-09	ASO 2013	-0.38	-0.32	0.0	0.05	-0.77	0.06	0.3	0.3	0.4
2013-10	SON 2013	-0.22	-0.14	-0.19	-0.15	-0.67	0.12	-0.1	-0.1	-0.2
2013-11	OND 2013	-0.23	-0.12	-0.19	-0.09	-0.64	0.14	0.7	1.3	0.0
2013-12	NDJ 2013	-0.27	-0.11	-0.24	-0.18	-0.01	-0.3	0.1	1.6	0.3
2014-01	DJF 2014	-0.43	-0.24	-0.6	-0.49	0.19	-0.32	1.4	-0.3	0.2
2014-02	JFM 2014	-0.38	-0.21	-0.97	-0.85	-0.92	-0.16	0.1	-0.9	-0.4
2014-03	FMA 2014	-0.15	-0.02	-0.47	-0.34	-0.79	0.05	-0.9	-0.6	-0.2
2014-04	MAM 2014	0.11	0.25	0.05	0.18	-0.35	0.32	0.8	-0.2	-0.0
2014-05	AMJ 2014	0.15	0.35	0.3	0.45	0.57	0.48	0.5	0.3	-0.5
2014-06	MJJ 2014	0.0	0.22	0.21	0.47	1.22	0.22	0.2	0.2	0.2
2014-07	JJA 2014	-0.14	0.07	-0.2	-0.03	1.36	0.04	-0.2	0.1	0.0
2014-08	JAS 2014	-0.09	0.07	-0.2	-0.01	1.02	0.28	-0.7	0.1	-0.8
2014-09	ASO 2014	0.06	0.24	0.16	0.31	0.74	0.53	-0.7	0.1	-0.1
2014-10	SON 2014	0.35	0.51	0.22	0.43	0.39	0.47	-0.6	0.1	-0.3
2014-11	OND 2014	0.54	0.7	0.58	0.75	0.74	0.73	-0.9	0.8	-0.6
2014-12	NDJ 2014	0.62	0.8	0.46	0.63	-0.11	0.75	-0.6	0.9	-0.1
2015-01	DJF 2015	0.54	0.73	0.34	0.51	0.06	0.7	-0.8	-0.2	-0.8
2015-02	JFM 2015	0.47	0.65	0.61	0.75	-0.57	0.96	0.2	-0.4	-0.5
2015-03	FMA 2015	0.59	0.72	0.33	0.44	0.13	0.94	-0.7	-1.8	-0.2
2015-04	MAM 2015	0.72	0.86	0.7	0.82	0.96	1.1	-0.0	-0.6	-0.6
2015-05	AMJ 2015	0.89	1.04	0.66	0.83	1.34	0.89	-0.7	-1.2	-1.2
2015-06	MJJ 2015	1.02	1.19	0.83	1.02	1.91	0.9	-0.6	-0.6	-0.6
2015-07	JJA 2015	1.27	1.44	0.94	1.26	2.48	0.89	-1.1	-1.5	-2.2
2015-08	JAS 2015	1.54	1.73	1.3	1.65	1.8	0.73	-1.4	-1.1	-1.3
2015-09	ASO 2015	1.8	2.02	1.35	1.79	2.26	0.79	-1.6	-1.3	-1.0

Month	Season	RONI	ONI	rel. Niño 3.4 (OISST)	Niño 3.4 (OISST)	Niño 1+2 (OISST)	Niño 4 (OISST)	SOI	W Pac trades	C Pac trades
2015-10	SON 2015	2.03	2.28	1.85	2.21	2.2	0.95	-1.7	-1.1	-2.3
2015-11	OND 2015	2.18	2.45	2.41	2.72	2.04	1.55	-0.5	0.0	-1.2
2015-12	NDJ 2015	2.25	2.59	2.0	2.39	2.1	1.4	-0.6	-0.1	-1.1
2016-01	DJF 2016	2.14	2.5	2.02	2.47	1.21	1.26	-2.2	-0.4	-2.2
2016-02	JFM 2016	1.81	2.21	1.92	2.23	1.01	1.37	-2.0	0.1	-0.5
2016-03	FMA 2016	1.3	1.69	1.19	1.61	1.11	1.2	-0.1	0.1	-0.2
2016-04	MAM 2016	0.65	1.11	0.58	0.98	0.45	0.75	-1.2	0.2	-0.2
2016-05	AMJ 2016	0.08	0.57	-0.25	0.27	0.44	0.56	0.4	0.7	0.1
2016-06	MJJ 2016	-0.43	0.09	-0.53	0.02	0.37	0.41	0.6	0.0	-0.4
2016-07	JJA 2016	-0.7	-0.19	-0.91	-0.42	0.39	0.15	0.4	0.3	-0.2
2016-08	JAS 2016	-0.81	-0.34	-0.93	-0.49	0.4	-0.01	0.7	0.1	0.3
2016-09	ASO 2016	-0.88	-0.42	-0.83	-0.43	0.59	-0.16	1.2	0.8	0.3
2016-10	SON 2016	-0.96	-0.51	-1.13	-0.7	0.48	-0.4	-0.3	0.7	0.2
2016-11	OND 2016	-0.96	-0.49	-1.04	-0.64	0.26	-0.41	-0.1	0.8	0.2
2016-12	NDJ 2016	-0.89	-0.37	-0.75	-0.33	0.42	-0.06	0.3	0.7	-0.0

RONI and ONI are three-month means centred on the month. SOI and trade-wind indices are standardized (negative = El Niño-like). OLR in W m^{-2} (negative = more convection). Heat content in °C vs 1981–2010; WWV in 10^{14} m^3; DMI in °C; PDO standardized.

Timeline

Date	Observation	Proposed mechanism	Source	Confidence
Jul 2013 – Feb 2014	ENSO-neutral, slightly cool (RONI −0.48 in JJA 2013, then −0.22 to −0.43 through DJF 2014); warm water volume positive throughout (+0.78 to +1.25 × 10^{14} m^3). A strong warm anomaly forms in the NE Pacific in winter	Recharged equatorial Pacific; the Blob from reduced ocean heat loss and weak cold advection under persistent high pressure.	[3] [7] [25]	High

Date	Observation	Proposed mechanism	Source	Confidence
	2013–14 (the 'Blob'), the largest there since at least the 1980s.			
Mid-2012 – mid-2014	TAO reporting declines sharply. In this analysis, daily equatorial wind records exist for only 33–41% of site-days from July 2013 to March 2014 and 36% in April–June 2014, recovering to 83–97% from October 2014 to September 2015.	Observing-system gap exactly during the 2014 onset attempt; 2014 wind-forced ocean studies must fill gaps.	[35] [19] [5]	High
Jan – Mar 2014	Two strong westerly bursts: reanalysis 24 Jan 2014 – 29 Jan 2014 (+9.5 m s^{-1} at 145°E) and 19 Feb 2014 – 5 Mar 2014 (+9.2 m s^{-1} at 155°E). Heat content east of the Date Line rises from −0.33 °C (Jan) to +1.60 °C (Mar); warm water volume peaks at +2.05 × 10^{14} m^3 in March; until April the warm pool is displaced east as at the onset of strong events; some models point to an extreme event.	Burst-forced downwelling Kelvin waves on a recharged basin, as in early 1997.	[6] [17] [16] [3]	High
Apr – Jun	No westerly wind events from April	Lack of spring bursts limits	[16] [6] [3]	High

Date	Observation	Proposed mechanism	Source	Confidence
2014	to June; the warm pool retreats to its usual position by July; RONI peaks at only +0.15 (AMJ) and falls to −0.14 (JJA).	eastern growth (contrast with 1997).		
Early 2014	Cold SST in the south-eastern Pacific and warm SST to the north (South Pacific meridional mode pattern); in forecast experiments about 40% of the peak-phase amplitude error comes from failing to predict the south-eastern cold anomalies.	Meridional SST gradient drives strong meridional rather than zonal winds near the equator.	[20]	Medium (one model)
Jun – Jul 2014	An unusually strong, basin-wide easterly wind burst in June; all major Niño indices fall to near normal; two separate warm centres persist. In the reanalysis the burst is broad rather than intense: the 150°E–100°W mean 1000 hPa anomaly reaches -1.94 m s^{-1} on 11 Jun 2014, the 5th-lowest day of the 1280 analysed, but no 5 m s^{-1} easterly	Easterly burst suppresses the Bjerknes feedback, and also prevents, then reverses, the discharge of equatorial heat content.	[17] [18] [6]	High (observation + model experiments)

Date	Observation	Proposed mechanism	Source	Confidence
	burst meets the burst definition.			
Aug – Dec 2014	Warm water volume rebuilds to +1.49 and +1.57 × 10^{14} m^3 (Aug–Sep); RONI rises to +0.62 (NDJ 2014–15); ONI ≥ +0.5 from SON 2014. The 2014 event 'failed to qualify as an El Niño by conventional definitions'.	Head start of equatorial heat content for 2015.	[3] [18]	High
2014 – 2015	The Blob persists for multiple years; PDO positive from 2014 (up to +1.86 in Dec 2014).	An enhanced second ocean–atmosphere coupled mode (Victoria mode, forced by a North Pacific Oscillation-like pattern) proposed as the bridge to the 2015 El Niño, triggered from the Northern Hemisphere.	[26] [27] [3]	Medium (one statistical study)
Mar 2015	The strongest early-2015 burst: reanalysis 4 Mar 2015 – 16 Mar 2015, +12.4 m s^{-1} at 167.5°E (140°–175°E), during an active MJO (OMI 2.03). TAO traces the downwelling Kelvin wave from 165°E on 11 Mar 2015 to 110°W on 12 Apr 2015 (32 days, about 3.4 m s^{-1}). Western-	Burst-forced Kelvin wave on a still-charged basin.	[6] [5] [3]	High

Date	Observation	Proposed mechanism	Source	Confidence
	Pacific trade index −1.8; central-Pacific OLR −18.8 W m^{-2}.			
Apr – Nov 2015	Four downwelling Kelvin waves; eastward current anomalies advect the warm pool east in spring and summer; a large equatorial freshwater anomaly, beyond anything in the 2004–2014 Argo record, raises steric height in the west and accelerates eastward flow at the salinity front.	Kelvin waves and advection initiate and enhance eastern-central warming.	[13] [32] [5]	High
May – Jul 2015	Coupling established: RONI AMJ +0.89, MJJ +1.02, JJA +1.27; SOI −0.6 to −1.1; Niño 1+2 +1.34, +1.91, +2.48 °C; public El Niño advisories in the first half of 2015.	Bjerknes feedback takes over.	[3] [12]	High
Jun – Oct 2015	Bursts keep coming during growth: 12 reanalysis bursts from late June to October, several centred east of the Date Line; calendar 2015 has 17, the most of any year in 1979–2016.	Bursts increasingly a response to the warming (state-dependent), reinforcing growth.	[6] [9] [22]	High (count); medium (interpretation)

Date	Observation	Proposed mechanism	Source	Confidence
Aug – Dec 2015	Growth to the peak: RONI SON +2.03, OND +2.18, NDJ +2.25 (ONI +2.59); Niño 4 reaches a record +1.55 °C (Nov) while Niño 1+2 stays near +2 °C; enhanced convection displaced about 20° west of 1982 and 1997; the eastern thermocline anomaly near South America only a third to a half of theirs; central-Pacific OLR −49.9 W m^{-2} in December.	Westward-centred warming: eastern-Pacific dynamics plus stronger central-Pacific (subtropically forced) dynamics.	[3] [13] [12] [15]	High
Nov 2015	An upwelling Kelvin wave emerges in early November, leading to a rapid decline; warm water volume turns negative in December (−0.39 × 10^{14} m^3).	Discharge begins.	[13] [3]	High
Jan – May 2016	Decay: RONI JFM +1.81 → AMJ +0.08; WWV −1.26 to −3.36 × 10^{14} m^3; warm SST lingers near the Date Line rather than retreating to the coast; demise in May 2016.	Discharge; stronger central-Pacific dynamics in decay.	[3] [15] [12]	High
Jun –	Weak La Niña (RONI −0.43 MJJ to −0.96	Negative IOD promoted the weak La Niña.	[3] [28]	High (observation); one model study (mechanism)

Date	Observation	Proposed mechanism	Source	Confidence
Dec 2016	SON/OND) with a strong negative Indian Ocean Dipole peaking in July–September 2016 (CPC DMI −0.89 °C in July).			

Confidence refers to the observation and its link to the stated mechanism for this event, not to general ENSO theory.

Evidence table

Factor	Observed conditions and timing	Proposed mechanism	Role in development	Evidence type	Confidence	Key sources
Recharge carried over from 2014	WWV positive from mid-2013; partial discharge in mid-2014 reversed after the easterly burst; WWV +0.31 to +1.25 × 10^{14} m^3 in Jan–May 2015 (against +3.12 in Apr 1997).	Retained heat content gave 2015 a head start.	Precondition	A−: observation + model analysis for this event	High	[3] [18] [7]
Easterly wind burst, June–July 2014	Strong basin-wide easterly anomaly; Niño indices fall; discharge prevented. Broad rather than intense in the reanalysis.	Suppressed 2014 growth while preserving heat for 2015.	Suppressor (2014) / indirect precondition (2015)	A: observation + model experiments	High	[17] [18] [6]
Westerly wind	3 reanalysis bursts in	Burst sequence	Trigger (2015); failed	A−: event observations	High	[6] [16] [19] [13]

Factor	Observed conditions and timing	Proposed mechanism	Role in development	Evidence type	Confidence	Key sources
events, 2014 and 2015	2014 (two in Jan–Mar, none in Apr–Jun) against 17 in 2015 (the most of any year 1979–2016; 1997 had 16); four downwelling Kelvin waves Apr–Nov 2015. A wind-forced ocean model reproduces 2014 and 2015 SST from bursts and easterly surges alone.	decides whether onset proceeds (2014 vs 2015 contrast).	trigger (2014)	+ ocean simulations		[21]
South Pacific meridional SST gradient (2014)	Cold south-east / warm north pattern in early 2014.	Meridional rather than zonal winds near the equator damp growth.	Suppressor (2014)	B/C: one forecast-model study	Medium	[20]
North Pacific (Blob, Victoria mode, meridional mode)	NE Pacific warm anomaly from winter 2013–14, multi-year; positive PDO 2014–16.	North Pacific atmosphere → subtropical SST and winds → equatorial westerlies; 2015 triggered from the Northern Hemisphere, unlike the Southern Hemisphere 'booster' found for	Possible precondition / trigger component	B/C: one coupled-mode statistical study for this event; general hindcast study	Medium–low	[27] [25] [23] [24]

Factor	Observed conditions and timing	Proposed mechanism	Role in development	Evidence type	Confidence	Key sources
		earlier super events.				
MJO	OMI active (≥ 1) on 3 Mar 2015 – 2 Apr 2015 and 29 May 2015 – 22 Jul 2015 (max 2.89), framing the March and late-June bursts; intraseasonal OLR variance at 160°E–150°W in Oct 2015 – Mar 2016 nearly twice that of Oct 1996 – Mar 1997. In a model, El Niño-like SST shifts the MJO onto the equator and strengthens it in the central Pacific, favouring bursts; March MJO alone has limited impact on ENSO in hindcasts.	MJO organises bursts; warming feeds back on the MJO.	Trigger timing / feedback	B: event indices + model experiments	Medium	[6] [22] [23] [10] [8]
Bjerknes feedback and central-Pacific dynamics	Coupled growth from May 2015; record Niño 4 and westward-	Eastern (thermocline) and central-Pacific (subtropically forced)	Amplifier / redirector of the pattern	B: event observations + one model comparison	Medium–high	[13] [15] [12] [34]

Factor	Observed conditions and timing	Proposed mechanism	Role in development	Evidence type	Confidence	Key sources
	shifted convection; eastern thermocline anomaly a third to a half of 1982 and 1997.	dynamics combined.				
Freshwater (salinity) anomaly 2015	Large equatorial freshwater anomaly in 2015, beyond the 2004–2014 Argo record.	Positive steric height in the west and eastward acceleration at the salinity front.	Amplifier (ocean dynamics)	B: event-specific budget	Medium	[32]
Seasonal cycle in termination	Southward shift of westerly anomalies in boreal winter is a key element of the termination of strong events.	SPCZ-related weakening of climatological winds south of the equator.	Termination mechanism	B/C: general model studies; 2015–16 not isolated in the sources read	Medium	[31] [30]
Indian Ocean	Weak positive DMI in Aug–Oct 2015 (+0.42 to +0.47 °C); strong negative IOD in mid-2016 promoting the weak La Niña.	IOD phase largely set by ENSO combination-mode forcing; the 2016 negative IOD developed mainly from Indian Ocean conditions.	Consequence (2015); influence on the transition (2016)	B: event-specific forecast experiments (2016); theory (combination mode)	Medium	[3] [28] [29]
Volcanic forcing	Tropical VEI 4 eruptions: Sinabung (Sep 2013, 3°N), Kelud (Feb 2014, 8°S) and	No event-specific study links them to the 2014–16 evolution.	Not supported (untested)	D	Low	[4]

Factor	Observed conditions and timing	Proposed mechanism	Role in development	Evidence type	Confidence	Key sources
	Wolf (May 2015, 0°); none larger.					
Background warming	RONI peak +2.25 vs ONI +2.59: the tropical-mean adjustment lowers the peak by 0.34 °C, the largest of the three events; the three strongest events are not separable within SST uncertainty; a review attributes the event's extremity partly to the warm 2014 conditions and long-term background warming.	The index adjustment removes tropical-mean warming; a statement about the index, not a quantitative attribution.	Background (amplitude partly attributed in a review)	C: review statement; index comparison	Medium	[2] [14] [11] [33]

Evidence grades: A = event-specific observation plus a model or heat-budget study isolating the effect; B = event-specific observation with a plausible mechanism and supporting studies; C = correlation, composite or general theory only; D = speculative, disputed, or not yet tested for this event.

Rainfall and tropospheric temperature (GPCP, NOAA STAR)

Rainfall in the usual El Niño impact regions, from GPCP v2.3 against 1991–2020. In 2015–16, 6 of the 8 regions had the usual El Niño sign. The most extreme seasons (top three since 1979) were: Central equatorial Pacific (SON, 317% of normal, 3rd wettest); Indonesia (Jul–Oct, 45% of normal, 1st driest). The same comparison across all four events is in Paper 4.

The tropical lower troposphere (NOAA STAR TLT, 20°S–20°N) warmed to +0.99 °C above 1991–2020 by Mar 2016, the usual lag of a few months behind the sea surface.

Region (season)	Usual El Niño signal	2015–16
Central equatorial Pacific (5°S–5°N, 170°E–140°W) (SON)	wetter	317% 3rd wettest of 47
Coastal Peru and Ecuador (12.5°S–2.5°N, 85°–77.5°W) (Dec–Apr)	wetter	91% 25th wettest of 47
Indonesia (10°S–5°N, 95°–145°E) (Jul–Oct)	drier	45% 1st driest of 47
Eastern Australia (40°–10°S, 140°–155°E) (Jun–Nov)	drier	75% 7th driest of 47
India (8°–30°N, 70°–90°E) (Jun–Sep)	drier	94% 12th driest of 47
Equatorial East Africa (5°S–5°N, 35°–50°E) (Oct–Dec)	wetter	107% 14th wettest of 47
Southern Africa (35°–15°S, 15°–40°E) (Dec–Mar)	drier	73% 4th driest of 47
California (32.5°–42.5°N, 125°–115°W) (Nov–Mar)	wetter	96% 22nd wettest of 47

GPCP v2.3 monthly precipitation (NOAA PSL), box means over land and ocean on the 2.5° grid; percentage of the 1991–2020 mean for the same months; rank among the 47 seasons 1979–2025 (wettest for regions that usually turn wet in El Niño, driest for those that usually turn dry). Data to Jun 2026. Seasons are the usual El Niño impact seasons, counted from the onset year (Dec–Apr = December of the onset year to April of the next).

Stage 5 verification log (24 October 2026)

Data. Every number in the text taken from the monthly table, the TAO analysis or the reanalysis and MJO diagnostics was recomputed from the source files by `tools/papers/verify_2015.py`: 48 checks, all passing. Diagnostic values are inserted by the build script from `data/papers/2015/diag_2015.json`, not typed by hand; the 1997 burst count from the extended run (16) matches Paper 2.

Literature. No full texts are held for this paper yet; every literature claim was checked against the published abstract recorded in the reference list. Studies listed as bibliographic records only are cited for nothing beyond their titles.

Corrections made while drafting. TAO data return is stated as a range per period rather than a single recovery figure; the October 2015 – March 2016 intraseasonal OLR variance is 'nearly twice' (ratio 1.83) rather than 'about twice' that of 1996–97; 16 uncited candidate references were removed from the list.

Open items. Full texts of Xue and Kumar (2017), Tseng et al. (2017), Gasparin and Roemmich (2016) and Lim and Hendon (2017) would allow numbers beyond the abstracts to be checked; island sea-level records could not be retrieved.

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Every DOI was checked against the Crossref registry; summaries rely on the published abstract unless noted. Full texts behind publisher paywalls were not read and are marked.

The four papers

Data snapshot: 24 October 2026. Independent, non-commercial analysis built from public sources (see About & method). Analysis prepared with Claude (Anthropic).